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The Editor-in-Chief
IEEE Access

Dear Editor,

We submit the manuscript "Coupling Authorization and Payment Layers in a PageRank Walk for Smart-Contract Reputation: A Pre-Specified, Commit-Gated Out-of-Window Test on an Ethereum Rollup" for consideration as a Research Article in *IEEE Access*.

Reputation scores for blockchain addresses are usually computed from token payments, although the same addresses are also linked by token allowances, the on-chain records by which an owner authorizes a spender. The manuscript treats the two relations as layers of one network and tests, out of window, whether a single PageRank walk that mixes them ranks smart contracts on the Arbitrum One rollup better than a walk over either layer alone. Two decision rules were committed to the authors' public repository before this study extracted the logs that judge them, the second eight seconds before its window was extracted, and the extraction code checks that the registration and configuration are committed without local changes and records their SHA-256 hashes before it runs.

Both rules failed, and the manuscript reports them as failed. The coupled walk ranked later approvals better than the payment walk, but ranked later payers worse by more than the registered margin and did not stay within the margin on the liquidation-free trading of contract traders. A revision of the amount weights, decided after these results were known, is reported as a re-analysis under the same rules and reached the same verdicts. The paper states what the commit-gated record does and does not guarantee (commit identifiers, file hashes and cloud job times are given), and calibrates known link-farm results to the authorization layer.

We believe the work matters to an international readership. Lending and margin protocols face pseudonymous counterparties and integrations, and on-chain reputation is a recurring proposal for screening them; the paper shows on public data that mixing the layers did not help, that a raw count of approving owners beat the authorization walk on later approvals, and how few approvals can inflate that walk. Because *IEEE Access* reviews submissions for technical soundness, it suits a study whose central findings are rules that were not met.

The manuscript is original, has not been published, and is not under consideration elsewhere. It is derived from the first author's doctoral research at the University of South Africa. A related manuscript by the same authors is being submitted at the same time to *PLOS ONE*. That paper compares the two edge types as separate scores against their raw degrees and on trader outcomes built from neither edge type; the present paper concerns the coupling operator, its two pre-specified rules and their record, and the manipulability of the authorization layer. The two papers share the data and cite each other. Both report the spender holdout coefficients of the raw degrees, EndorseRank and AWP with their degree contrasts, the closed form of the authorization walk and the solver runtimes, and these reference rows are attributed to the companion paper here. The allowance-against-transfer comparison on trader outcomes appears only in the companion; on that trader cohort the present paper reports only the contrasts of its coupled walk that its own registered rule names. We upload the companion manuscript as confidential material for the reviewers. The study uses only public blockchain and price data and involves no human participants. The code, configuration, pre-specification documents, result summaries and the manuscript's scripts are public at `https://github.com/abitocodes/phd_works` and `https://github.com/abitocodes/phd-dissertation-draft-en`. The research received no external funding. The authors declare no competing interests, and all authors have approved the submission.

Yours sincerely,

Taehong Kwon (corresponding author), on behalf of Ernest Mnkandla, Donatien Koulla Moulla and David Sena Attipoe